Learning Augmentation Weights Through One Classifier Update in Corruption-Robust Continual Learning

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Abstract

How does the share of the original image in an augmented view affect corruption robustness in continual learning? We compare chain-aware mixing and a recurrent MetaMix baseline at five original-image weights on Split-CIFAR-10. Across three seeds, classifier InfoNCE improves mean corrupted class-incremental accuracy over Jensen–Shannon (JS) consistency at all five weights for chain-aware mixing and at four for MetaMix. At the strongest tested augmentation ($m = 0.1$), the two InfoNCE classifiers reach 43.89% and 41.66%, respectively. At $m = 0.3$ their order reverses (42.28% and 42.51%). The AugMix reference leads on corrupted task-incremental accuracy at 84.15%. The results link classifier consistency to mixture strength and show why conclusions about mixer design depend on which metric and strength are compared.

1. Introduction

An augmentation mixer can reduce disagreement between a clean image and its transformed views by assigning more weight to the original. At the extreme, all views coincide. This presents a difficulty when prediction consistency is used to train the mixer: a smaller loss can reflect an easier view even when the classifier has learned little about corruptions. In continual learning, the choice of views also affects the update shared by current examples and examples replayed from earlier tasks.

AugMix draws and blends short stochastic augmentation chains [3]. MetaMix adapts mixture weights using a recurrent summary of the learner and splits training operations into pseudo-seen and pseudo-unseen pools [4]. Specifically, MetaMix baseline uses an LSTM ϕ conditioned on current and replay features, replay loss and gradient alignment. Given the current task's mini batch (x_t, y_t) at time step t, and a labeled replay batch (x_r, y_r), we combine them into (x_b, y_b). From the set of training augmentations, to simulate the real scenario in which we are not accessible to the test augmentations, we construct two disjoint set of augmentations: pseudo-seen pool S_t and pseudo-unseen pool U_t. S_t produces two mixed views x'_{b1}, x'_{b2} of this batch; the disjoint pseudo-unseen pool U_t produces $\hat{x}_{b1}, \hat{x}_{b2}$. For either pool, a mixed view of one image x has the form

$$\tilde{x} = mx + (1 - m) \sum_{k=1}^{3} w_k a_k(x), \qquad \sum_{k=1}^{3} w_k = 1, \quad (1)$$

where m is the original-image weight, a_k is a sampled chain from S_t or U_t, and $w_k \geq 0$ is its weight. We write θ for classifier parameters and ϕ for LSTM (aka mixer) parameters. The paper aims to learn a mixer ϕ that can guide the classifier to give similar representation between x and $\tilde{x}$.

However, since m is a learnable parameter, we empirically verified that during training m approaches 1 which consequently make the mixed image $\tilde{x}$ more similar the original image. Also, as the mixer doesn't receive any information about the current pseudo-seen chains sampled from S_t, the mixer will treat every chain equally and overlook the chains that benefits the current task training.

Based on this we study a modification with two linked choices. First, the mixer observes how each sampled pseudo-seen chain changes the image's representation and prediction. Next, since the mixer can observe the effect of the augmentations to select chain, instead of using LSTM to mix the unseen chains as in the previous work, we generate independent random weights to mix pseudo-unseen views on which the classifier update is scored. This design lets the mixer influence the outer loss only through its choice of training views.

Our contributions are:

- We introduce a chain-aware mixer that combines temporal learner context with each sampled chain's feature and prediction changes to assign weights separately for each image and pseudo-seen view.

- We train the mixer through one differentiable classifier update, using independently sampled pseudo-unseen mixture weights. This makes the mixer's outer-loss gradient depend entirely on how its training mixtures affect the classifier update.
- We compare mixer design, classifier consistency, and augmentation strength on Split-CIFAR-10 across five fixed original-image weights and three seeds, including MetaMix and AugMix baselines. The results show that the relative performance of learned mixing depends on mixture strength and evaluation metric.

2. Training Objective Design

What the mixer observes. Before the classifier update, we compute mean cross-entropy on the clean current batch and the clean labeled replay batch. Let g_t and g_r be their gradients with respect to the classifier parameters, and let L_r be the replay cross-entropy. Each LSTM input concatenates these batch statistics with individual pooled features:

$$I_i = [g_t^\top g_r,\ L_r,\ e_{r,i},\ e_{t,i}] \in \mathbb{R}^{258}. \tag{2}$$

Here $e_{t,i}$ and $e_{r,i}$ are 128-dimensional features from clean current and replay images. The gradient dot product measures whether their batch losses favor locally compatible updates: a positive value indicates that a sufficiently small step along the negative current gradient would decrease replay loss to first order, whereas a negative value indicates conflict. Replay loss summarizes how well the classifier currently fits remembered examples, and the feature vectors provide image-specific context.

The MetaMix baseline maps the LSTM hidden state to three chain weights shared across views and operation pools for each example. One variant also learns m, while the other uses the prescribed sweep value. Because the LSTM context contains no sampled-chain information, resampling chains at fixed context and recurrent state leaves these weights unchanged.

The chain-aware mixer uses the same LSTM context and the prescribed m. For each seen chain, it concatenates the change in pooled features with the change in class probabilities relative to the clean input. We then concatenate this augmentation descriptor with LSTM hidden state from the previous step and mapped to a scalar. A softmax over 3 scores (correspond to 3 sampled chains) supplies weights separately for each example and seen view. For each pseudo-unseen view we draw $w \sim \mathrm{Dirichlet}(1, 1, 1)$ independently of the mixer and keep the draw fixed throughout the batch's mixer updates.

What the classifier learns. Write $\ell_\theta(x, y) = \mathrm{CE}(f_\theta(x), y)$ for mean batch cross-entropy, where f_θ returns logits. A separate replay draw (x_d, z_d) supplies stored logits for distillation. Our DER++ loss [1] is

$$L_{\mathrm{DER++}}(\theta) = \ell_\theta(x_t, y_t) + 0.5\,\ell_\theta(x_r, y_r) + 0.2\,\mathrm{MSE}(f_\theta(x_d), z_d). \tag{3}$$

MSE averages over examples and logit coordinates. Replay terms are zero when memory is empty. The classifier objective is

$$L_S(\theta, \phi) = L_{\mathrm{DER++}}(\theta) + C_\theta(x_b, x'_{b1}, x'_{b2}). \tag{4}$$

Here C_θ is either Jensen-Shannon divergence or class-aware InfoNCE; the latter also uses the labels y_b. The dependence on ϕ enters through the mixed views.

For three aligned batches v_0, v_1, v_2 of size n, let p_{ij} denote the classifier's probability vector for example i in view v_j, and let $\bar{p}_i = (p_{i0} + p_{i1} + p_{i2})/3$. We use

$$\mathcal{L}_{\mathrm{JS},\theta}(v_0, v_1, v_2) = \frac{1}{3n} \sum_{i=1}^{n} \sum_{j=0}^{2} \mathrm{KL}(p_{ij} \,\|\, \bar{p}_i). \tag{5}$$

Beside JS divergence which encourage the consistency in the probability distributions, we also try InfoNCE which is a stronger regularization that directly constraints the feature similarities. Each image contributes three views: the original image and its two pseudo-seen augmented mixtures. We treat each view in turn as an anchor and use the other two views of that same image as positive examples. Index the $3n$ views by a and let u_a be the normalized pooled feature. The positive set P_a contains these two same-image views, while the negative set N_a contains all views of images with different class labels. The anchor itself and views of other images with the same class label are excluded from both sets. At temperature $\tau = 0.1$,

$$\mathcal{L}_{\mathrm{NCE},\theta} = -\frac{1}{3n} \sum_{a} \frac{1}{2} \sum_{b \in P_a} \log \frac{\exp(u_a^\top u_b/\tau)}{\sum_{c \in P_a \cup N_a} \exp(u_a^\top u_c/\tau)}. \tag{6}$$

How the mixer guides representation learning. The mixer chooses the pseudo-seen training views, which determine the consistency gradient used to update the classifier. We train the mixer by asking whether that update improves classification and prediction consistency on independently constructed pseudo-unseen views. After one virtual classifier step of size α, we evaluate

$$\theta^+(\phi) = \theta - \alpha \nabla_\theta L_S(\theta, \phi), \tag{7}$$

$$L_U(\theta^+, \phi) = \ell_{\theta^+}(x_b, y_b) + \mathcal{L}_{\mathrm{JS},\theta^+}(x_b, \hat{x}_{b1}, \hat{x}_{b2}). \tag{8}$$

Clean cross-entropy rewards retaining class-discriminative information, while pseudo-unseen JS rewards predictions that remain consistent under transformations outside the current pseudo-seen pool. The mixer minimizes their sum after

the classifier step, $J(\phi) = L_U(\theta^+(\phi), \phi)$, so its feedback concerns the effect of its mixtures on learning.

Holding the sampled batch, incoming state and outer BatchNorm buffers fixed, the chain rule gives

$$\nabla_\phi J = (D_\phi \theta^+)^\top \nabla_{\theta^+} L_U + \partial_\phi L_U. \tag{9}$$

The first term links a change in mixture weights to a change in the classifier update and, through that update, to the outer loss. The second term measures how the mixer changes the outer views while holding the virtual classifier fixed. For our chain-aware mixer, the pseudo-unseen mixture weights are sampled independently of ϕ and m is prescribed, so the second term vanishes. Consequently, the mixer can lower the outer loss only by changing the classifier step taken on its pseudo-seen mixtures. A direction $\delta\phi$ is locally favorable when

$$\left\langle \nabla_{\theta^+} L_U, (D_\phi \theta^+)\delta\phi \right\rangle < 0. \tag{10}$$

This expresses the intended guidance: favor changes in the training mixtures whose resulting classifier update reduces the combined classification and pseudo-unseen consistency loss.

3. Experimental Protocol

We compare the two mixers at matched original-image weights $m \in \{0.1, 0.3, 0.5, 0.7, 0.9\}$. At each weight we cross mixer choice with classifier consistency. Table 1 reports all four evaluation metrics so the reader can compare mixers within a row and follow each mixer as augmentation strength changes.

The 288,746-parameter SmallCNN learns five successive CIFAR-10 class pairs, $(0, 1)$ through $(8, 9)$, with ten epochs per task and a 500-example reservoir. Each batch has up to 32 current examples and up to 32 labeled replay examples. The classifier and mixer use constant SGD learning rates of 0.003 and 0.005. Learned mixers receive three SGD updates per mini-batch, while the classifier receives one committed update; the random AugMix references have no mixer updates. Each run processes 500,000 current-image presentations in 15,650 classifier updates.

Final evaluation uses 10,000 clean test images and 15 CIFAR-10-C corruptions at five severities [2]. Each corruption and severity receives equal weight. Class-IL predicts among all ten outputs. Task-IL restricts prediction to the true class pair. The pseudo-unseen operations used in training are sampled training transformations. CIFAR-10-C supplies only the test measurements.

References and aggregation. DER++ with AugMix samples two views from the full nine-operation pool, with independent $m \sim \mathrm{Beta}(1, 1)$ and chain weights drawn from $\mathrm{Dirichlet}(1, 1, 1)$. We evaluate its original JS coefficient of 12 and a coefficient of one matched to the JS classifiers. Both use the same replay and classifier update budgets. Each mean gives equal weight to seeds 17, 23, and 43. We report sample SD across seeds in Table 5 and compute differences from the displayed means.

4. Results

Mixture strength across four metrics. For chain-aware InfoNCE all four means decline as m increases (Table 1). The largest class-IL difference occurs on corruptions: 43.89% at $m = 0.1$ versus 34.18% at $m = 0.9$. MetaMix InfoNCE reaches its highest class-IL means at $m = 0.3$ and its highest task-IL means at $m = 0.1$. With JS, the highest corrupted class-IL mean occurs at $m = 0.1$ for both mixers, although intermediate values give a less regular ordering.

What the chain-aware design adds. With classifier InfoNCE and $m = 0.1$, chain-aware leads on all four metrics. At $m = 0.3$ MetaMix leads on corrupted class-IL by 0.23 points, while chain-aware leads on the other three metrics. At $m = 0.5$ their clean task-IL means coincide to two decimals. The comparison changes both seen-chain scoring and pseudo-unseen mixture construction.

Interaction with classifier consistency. Table 2 holds the mixer and mixture strength fixed when comparing classifier losses. Chain-aware gains 5.69 and 5.95 points on corrupted class-IL from InfoNCE at $m = 0.1$ and 0.3, compared with 0.93 at $m = 0.5$. MetaMix gains 8.20 points at $m = 0.3$ but loses 1.74 at $m = 0.9$. The gains also depend on task information: at $m = 0.7$ the chain-aware task-IL gains are only 0.31 points clean and 0.01 corrupted.

		Class-IL		Task-IL	
Mixer	m	Clean	Corrupt	Clean	Corrupt
Chain-aware	0.1	+7.88	+5.69	+2.41	+2.76
	0.3	+7.66	+5.95	+0.31	+0.65
	0.5	+1.67	+0.93	+0.86	+1.78
	0.7	+3.14	+1.76	+0.31	+0.01
	0.9	+0.46	+0.77	+0.97	+1.79
MetaMix	0.1	+3.56	+2.43	+1.18	+1.29
	0.3	+9.36	+8.20	+0.82	+2.03
	0.5	+2.20	+0.89	+0.23	+0.96
	0.7	+3.94	+1.94	+0.95	+0.52
	0.9	−1.59	−1.74	−0.02	+0.31

Table 2. Change in mean accuracy (percentage points) when the classifier switches from JS to InfoNCE, holding the mixer and m fixed.

Learned and random mixture strength. Tables 1 and 3 show how the fixed settings compare with learned m and random AugMix. Within MetaMix with classifier InfoNCE, corrupted class-IL is 35.26% with learned m, 42.51% at

Loss	m	Clean class-IL		Corrupt class-IL		Clean task-IL		Corrupt task-IL	
		CA	MM	CA	MM	CA	MM	CA	MM
JS	0.1	43.99	**45.69**	38.20	**39.23**	86.05	**86.56**	80.48	**80.88**
	0.3	**42.61**	40.78	**36.33**	34.31	**87.56**	86.61	**81.29**	79.82
	0.5	**44.59**	43.51	**36.56**	36.38	86.26	**86.89**	79.14	**80.05**
	0.7	**42.94**	41.69	**34.96**	34.26	**86.65**	86.12	**79.63**	79.07
	0.9	42.87	**45.60**	33.41	**36.34**	85.58	**86.55**	77.64	**78.88**
InfoNCE	0.1	**51.87**	49.25	**43.89**	41.66	**88.46**	87.74	**83.24**	82.17
	0.3	**50.27**	50.14	42.28	**42.51**	**87.87**	87.43	**81.94**	81.85
	0.5	**46.26**	45.71	**37.49**	37.27	**87.12**	**87.12**	80.92	**81.01**
	0.7	**46.08**	45.63	**36.72**	36.20	86.96	**87.07**	**79.64**	79.59
	0.9	43.33	**44.01**	34.18	**34.60**	**86.55**	86.53	**79.43**	79.19

Table 1. Mean final accuracy (%) over seeds 17, 23, and 43. Column pairs compare chain-aware (CA) and MetaMix (MM) at the same original-image weight and classifier consistency loss. For each loss and m, bold marks the higher CA or MM mean within each metric pair. Ties at the displayed precision are both bold.

fixed $m = 0.3$, and 34.60% at fixed $m = 0.9$. Thus the value chosen matters even when m is fixed. AugMix with its original JS coefficient reaches the highest corrupted task-IL mean (84.15%). Reducing that coefficient to one raises clean class-IL from 45.94% to 48.99%, while corrupted class-IL changes from 40.00% to 39.68%. At the matched coefficient AugMix exceeds both mixers' best observed JS means on corrupted class-IL.

		Class-IL		Task-IL	
Mixer	Loss	Clean	Corrupt	Clean	Corrupt
MetaMix, learned m	JS	46.16	36.43	86.65	79.83
MetaMix, learned m	InfoNCE	45.48	35.26	86.51	79.14
DER++ + AugMix	JS	**48.99**	39.68	87.92	81.81

Table 3. Reference methods with learned or sampled original-image weight. Entries are three-seed mean accuracies (%). Bold marks the highest mean in each metric among these reference methods.

Which corruptions account for the differences? Tables 6 and 7 give the results for each corruption. With InfoNCE and $m = 0.5$, chain-aware mixing exceeds MetaMix on 13 of 15 class-IL corruption means. The largest advantage is on pixelation (39.74% versus 39.10%). MetaMix leads on impulse noise and contrast. Changing the classifier loss within the chain-aware mixer produces larger differences: switching from JS to InfoNCE raises impulse-noise class-IL by 6.63 points but lowers fog class-IL by 3.57 points. When the true task is supplied, the same switch raises accuracy on 14 corruptions, with motion blur the exception (82.33% with JS and 81.72% with InfoNCE).

How concentrated are the learned weights? At $m = 0.1$, the augmentation chains contribute 90% to the mixed images $\tilde{x}$. For InfoNCE, the most heavily weighted chain receives 38.24% of that share on average with chain-aware mixing and 34.37% with MetaMix (Table 4).

Mixer	Loss	Max. weight
Chain-aware	JS	.3603
Chain-aware	InfoNCE	.3824
MetaMix	JS	.3463
MetaMix	InfoNCE	.3437

Table 4. Final-epoch weights at $m = 0.1$, averaged across seeds. The largest of the three weights is taken before averaging. Uniform weights give a maximum of $1/3$ and entropy $\ln 3$.

5. Limitations and Conclusion

The most consistent gain on corrupted class-IL is from classifier InfoNCE at stronger augmentation. Chain-aware InfoNCE attains the highest observed mean on this metric, while the advantage over MetaMix changes sign across matched strengths. The maximum is selected from five settings on the reported evaluation. With three seeds, dispersion is substantial: at $m = 0.1$ the two InfoNCE corrupted class-IL means are 43.89 ± 5.68 and 41.66 ± 7.68 (sample SD). Isolating the effect of candidate descriptors would require holding outer-view sampling and per-view weight sharing fixed. The per-corruption breakdown below covers the original six configurations; the concentration measurements cover $m = 0.1$.

Reproducibility. The repository includes the three-seed reporting script, the one-step trainer, and the PBS jobs for the fixed-strength sweep and AugMix coefficient comparison.

References

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[2] Dan Hendrycks and Thomas Dietterich. Benchmarking neural network robustness to common corruptions and perturbations. In *International Conference on Learning Representations*, 2019.

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A. Variation across training seeds

Mixer and loss	m	Clean class-IL	Corrupt class-IL	Clean task-IL	Corrupt task-IL
CA JS	0.1	43.99 ± 8.64	38.20 ± 6.12	86.05 ± 2.84	80.48 ± 1.92
	0.3	42.61 ± 5.12	36.33 ± 3.76	87.56 ± 1.01	81.29 ± 1.19
	0.5	44.59 ± 3.42	36.56 ± 3.15	86.26 ± 1.99	79.14 ± 1.77
	0.7	42.94 ± 7.54	34.96 ± 4.90	86.65 ± 1.00	79.63 ± 0.97
	0.9	42.87 ± 9.60	33.41 ± 7.11	85.58 ± 1.18	77.64 ± 1.91
CA InfoNCE	0.1	51.87 ± 7.29	43.89 ± 5.68	88.46 ± 0.72	83.24 ± 0.62
	0.3	50.27 ± 4.64	42.28 ± 3.03	87.87 ± 0.53	81.94 ± 0.29
	0.5	46.26 ± 7.46	37.49 ± 5.95	87.12 ± 1.47	80.92 ± 1.31
	0.7	46.08 ± 4.40	36.72 ± 3.01	86.96 ± 0.68	79.64 ± 1.22
	0.9	43.33 ± 6.16	34.18 ± 4.78	86.55 ± 0.63	79.43 ± 0.93
MM JS	0.1	45.69 ± 7.10	39.23 ± 4.69	86.56 ± 1.79	80.88 ± 1.08
	0.3	40.78 ± 6.02	34.31 ± 5.57	86.61 ± 2.68	79.82 ± 3.24
	0.5	43.51 ± 4.55	36.38 ± 3.30	86.89 ± 0.94	80.05 ± 0.62
	0.7	41.69 ± 4.83	34.26 ± 2.55	86.12 ± 1.35	79.07 ± 0.74
	0.9	45.60 ± 7.86	36.34 ± 4.37	86.55 ± 1.09	78.88 ± 0.86
MM InfoNCE	0.1	49.25 ± 9.09	41.66 ± 7.68	87.74 ± 1.88	82.17 ± 1.92
	0.3	50.14 ± 5.61	42.51 ± 3.66	87.43 ± 1.42	81.85 ± 1.15
	0.5	45.71 ± 8.40	37.27 ± 6.33	87.12 ± 1.80	81.01 ± 1.41
	0.7	45.63 ± 5.83	36.20 ± 3.97	87.07 ± 0.76	79.59 ± 1.31
	0.9	44.01 ± 5.66	34.60 ± 4.26	86.53 ± 0.51	79.19 ± 0.77

Table 5. Fixed-weight sweep, mean accuracy (%) $\pm$ sample SD over three seeds.

B. CIFAR-10-C accuracy by corruption

Each entry averages the five severity levels within a seed and then takes the arithmetic mean across seeds 17, 23 and 43. Overall gives equal weight to each of the 15 corruptions. Class-IL predicts among all ten classes. Task-IL predicts within the true class pair.

	Chain-aware, $m = 0.5$		MetaMix, $m = 0.5$		MetaMix, learned m	
Corruption	JS	InfoNCE	JS	InfoNCE	JS	InfoNCE
Gaussian Noise	28.76	**33.76**	30.49	33.61	29.49	30.42
Shot Noise	32.53	**36.57**	33.82	36.38	32.92	33.06
Impulse Noise	22.97	29.60	25.64	**29.70**	25.64	27.40
Defocus Blur	**44.85**	43.99	44.19	43.57	42.32	40.62
Glass Blur	34.00	**36.04**	34.17	36.01	35.28	34.27
Motion Blur	**41.58**	38.66	40.59	38.50	38.26	34.97
Zoom Blur	**44.66**	42.77	44.21	42.58	40.41	38.72
Snow	33.57	**36.51**	33.41	36.27	35.99	35.09
Frost	31.30	31.09	30.49	31.01	**32.48**	30.39
Fog	**39.47**	35.90	37.97	35.81	37.59	34.91
Brightness	39.68	40.77	38.47	40.34	**42.22**	40.50
Contrast	**30.16**	28.86	29.45	28.99	29.37	27.61
Elastic Transform	**44.42**	43.45	43.36	42.96	42.78	41.28
Pixelate	37.71	**39.74**	37.20	39.10	37.96	36.84
JPEG Compression	42.79	**44.68**	42.26	44.17	43.75	42.88
Overall	36.56	**37.49**	36.38	37.27	36.43	35.26

Table 6. Class-IL final accuracy (%) by corruption and classifier consistency. Bold marks the highest displayed mean in each row.

	Chain-aware, $m = 0.5$		MetaMix, $m = 0.5$		MetaMix, learned m	
Corruption	JS	InfoNCE	JS	InfoNCE	JS	InfoNCE
Gaussian Noise	73.07	77.64	75.04	**77.69**	74.89	73.52
Shot Noise	76.24	80.28	77.99	**80.56**	77.69	76.28
Impulse Noise	71.88	76.32	73.34	**76.40**	73.06	72.47
Defocus Blur	84.20	84.62	84.76	**84.79**	83.76	83.58
Glass Blur	73.70	77.05	75.27	**77.34**	75.58	74.83
Motion Blur	82.33	81.72	**83.09**	82.12	81.86	80.39
Zoom Blur	83.15	83.18	**83.75**	83.39	82.30	82.18
Snow	78.97	**81.59**	79.92	81.55	79.90	80.89
Frost	78.34	**80.09**	78.96	**80.09**	79.31	79.12
Fog	80.01	80.95	80.07	**81.00**	80.57	79.14
Brightness	83.97	**84.71**	84.60	84.54	84.56	84.43
Contrast	70.05	70.82	70.28	70.71	**71.21**	68.55
Elastic Transform	83.42	84.15	84.12	**84.36**	83.61	83.65
Pixelate	83.92	**85.66**	84.60	85.55	84.50	83.69
JPEG Compression	83.90	85.01	84.98	**85.10**	84.69	84.36
Overall	79.14	80.92	80.05	**81.01**	79.83	79.14

Table 7. Task-IL final accuracy (%) by corruption and classifier consistency. Bold marks the highest displayed mean in each row.